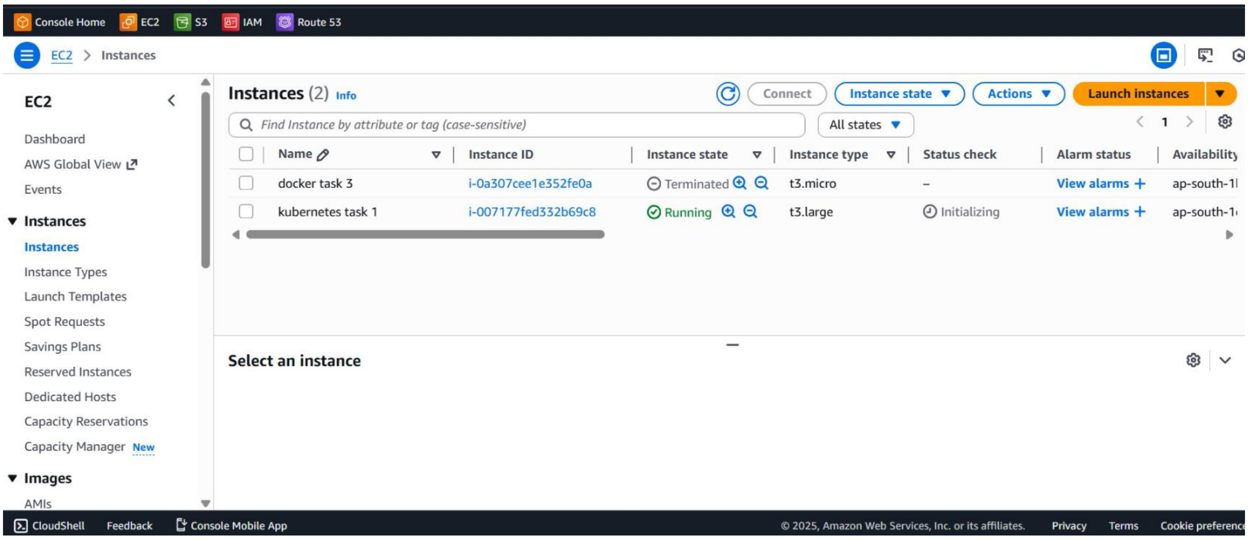
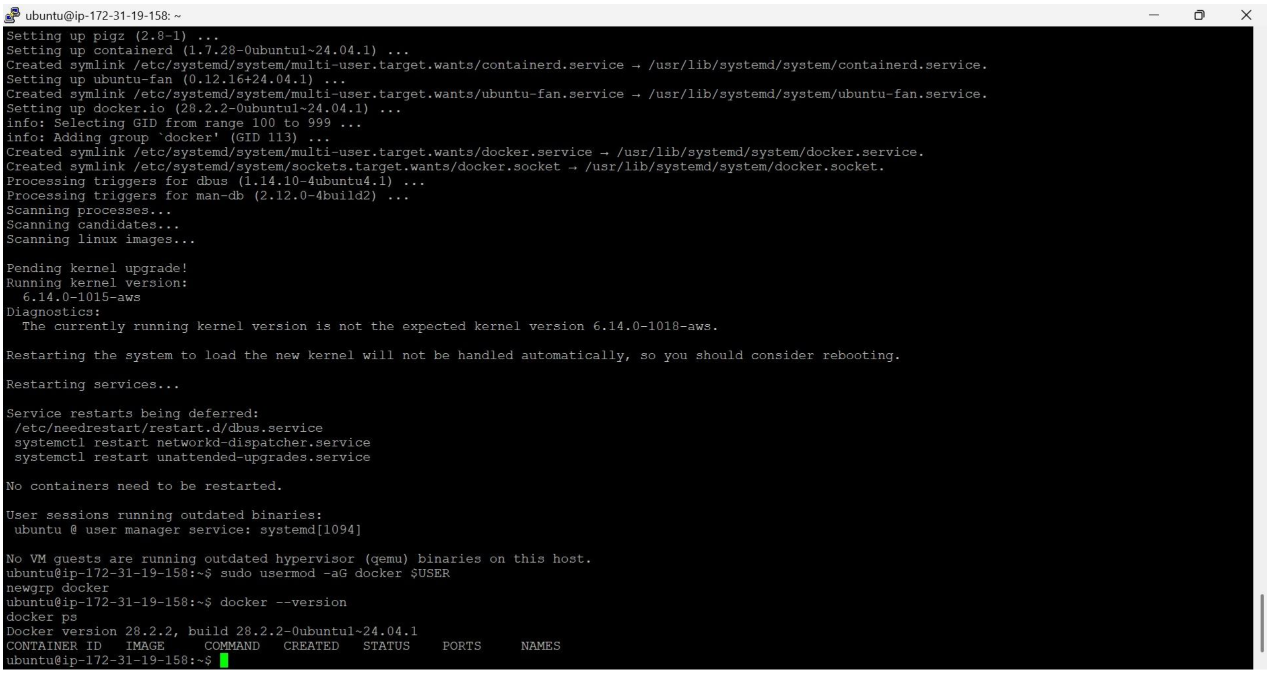
Kubernetes Task

Task Description:

Setup minikube at your local and explore creating namespaces (Go through official documentation).

<u>Output:</u>





ubuntu@ip-172-31-19-158: ~

```
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.14.0-1015-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.14.0-1018-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ user manager service: systemd[1094]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-19-158:~$ sudo usermod -aG docker $USER
newgrp docker
ubuntu@ip-172-31-19-158:~$ docker --version
docker ps
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-19-158:~$ curl -LO https://storage.googleapis.com/kubernetes-release/release/`curl -s \
https://storage.googleapis.com/kubernetes-release/release/stable.txt`/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0   9.9M      0  0:00:05  0:00:05 --:--:-- 12.8M
ubuntu@ip-172-31-19-158:~$ chmod +x kubectl
sudo mv kubectl /usr/local/bin/
ubuntu@ip-172-31-19-158:~$ kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
ubuntu@ip-172-31-19-158:~$
```

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  The currently running kernel version is not the expected kernel version 6.14.0-1018-aws.

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Client Version: v1.31.0
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ubuntu@ip-172-31-19-158:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
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100  133M  100  133M    0     0  12.6M      0  0:00:10  0:00:10 --:--:-- 15.6M
ubuntu@ip-172-31-19-158:~$ chmod +x minikube-linux-amd64
sudo mv minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@ip-172-31-19-158:~$ minikube version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
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commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
ubuntu@ip-172-31-19-158:~$ minikube start --driver=docker
* minikube v1.37.0 on Ubuntu 24.04
* Using the docker driver based on user configuration
* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.48 ...
* Downloading Kubernetes v1.34.0 preload ...
    > gcr.io/k8s-minikube/kicbase...:  488.51 MiB / 488.52 MiB  100.00% 49.62 M
    > preloaded-images-k8s-v18-v1...:  337.07 MiB / 337.07 MiB  100.00% 14.94 M
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  - Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-19-158:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

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NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   60s   v1.34.0
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ubuntu@ip-172-31-19-158:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   105s
kube-node-lease   Active   105s
kube-public       Active   105s
kube-system       Active   105s
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namespace/dev created
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namespace/test created
ubuntu@ip-172-31-19-158:~$ kubectl get ns
NAME              STATUS   AGE
default           Active   3m21s
dev               Active   52s
kube-node-lease   Active   3m21s
kube-public       Active   3m21s
kube-system       Active   3m21s
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```
ubuntu@ip-172-31-19-158:~$ kubectl run nginx-dev --image=nginx --namespace=dev
pod/nginx-dev created
ubuntu@ip-172-31-19-158:~$ kubectl get pods -n dev
NAME        READY   STATUS              RESTARTS   AGE
nginx-dev   0/1     ContainerCreating   0          7s
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pod/nginx-dev created
ubuntu@ip-172-31-19-158:~$ kubectl get pods -n dev
NAME        READY   STATUS              RESTARTS   AGE
nginx-dev   0/1     ContainerCreating   0          7s
ubuntu@ip-172-31-19-158:~$ kubectl describe namespace dev
Name:         dev
Labels:       kubernetes.io/metadata.name=dev
Annotations:  <none>
Status:       Active

No resource quota.

No LimitRange resource.
ubuntu@ip-172-31-19-158:~$
```